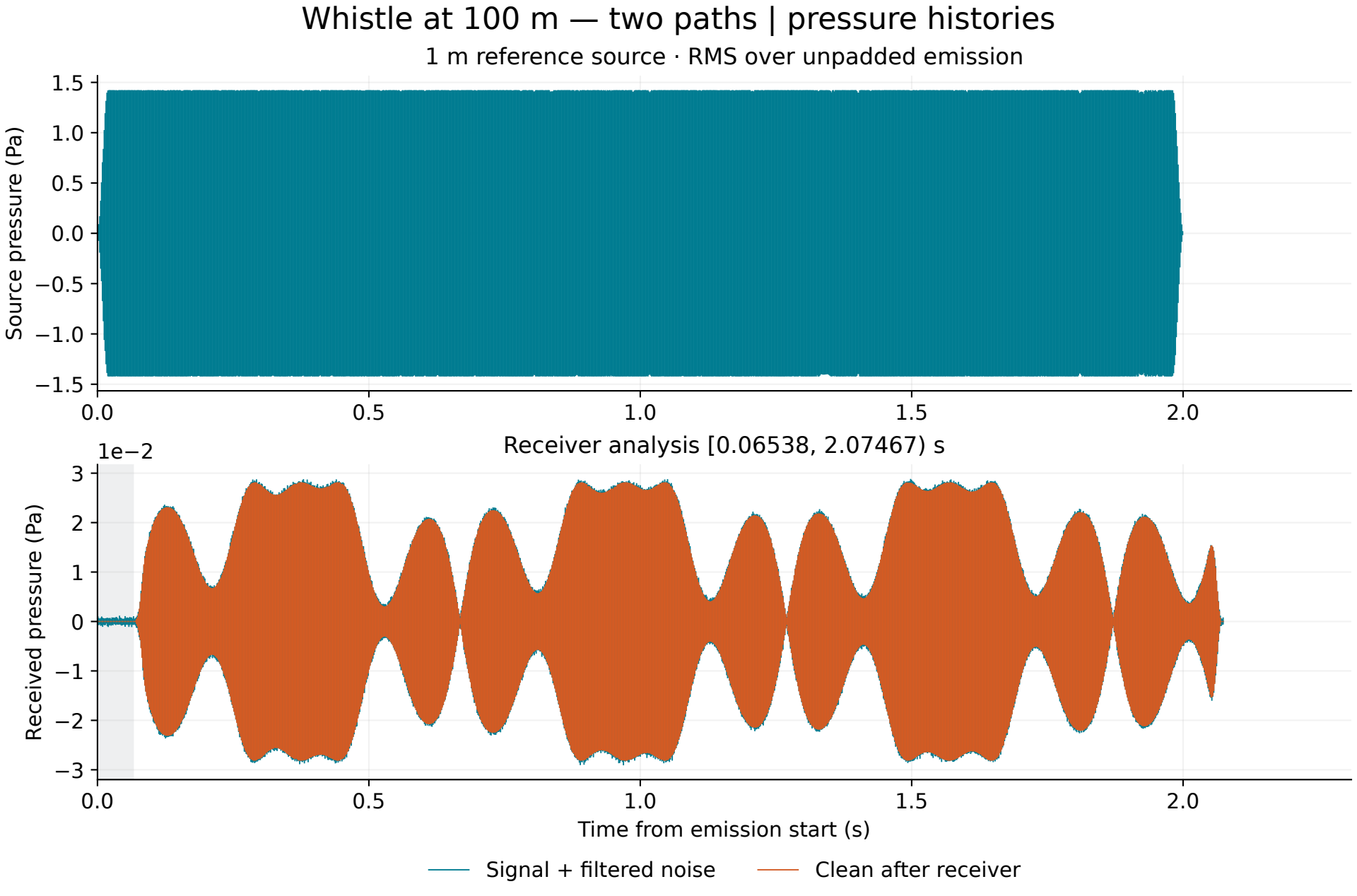
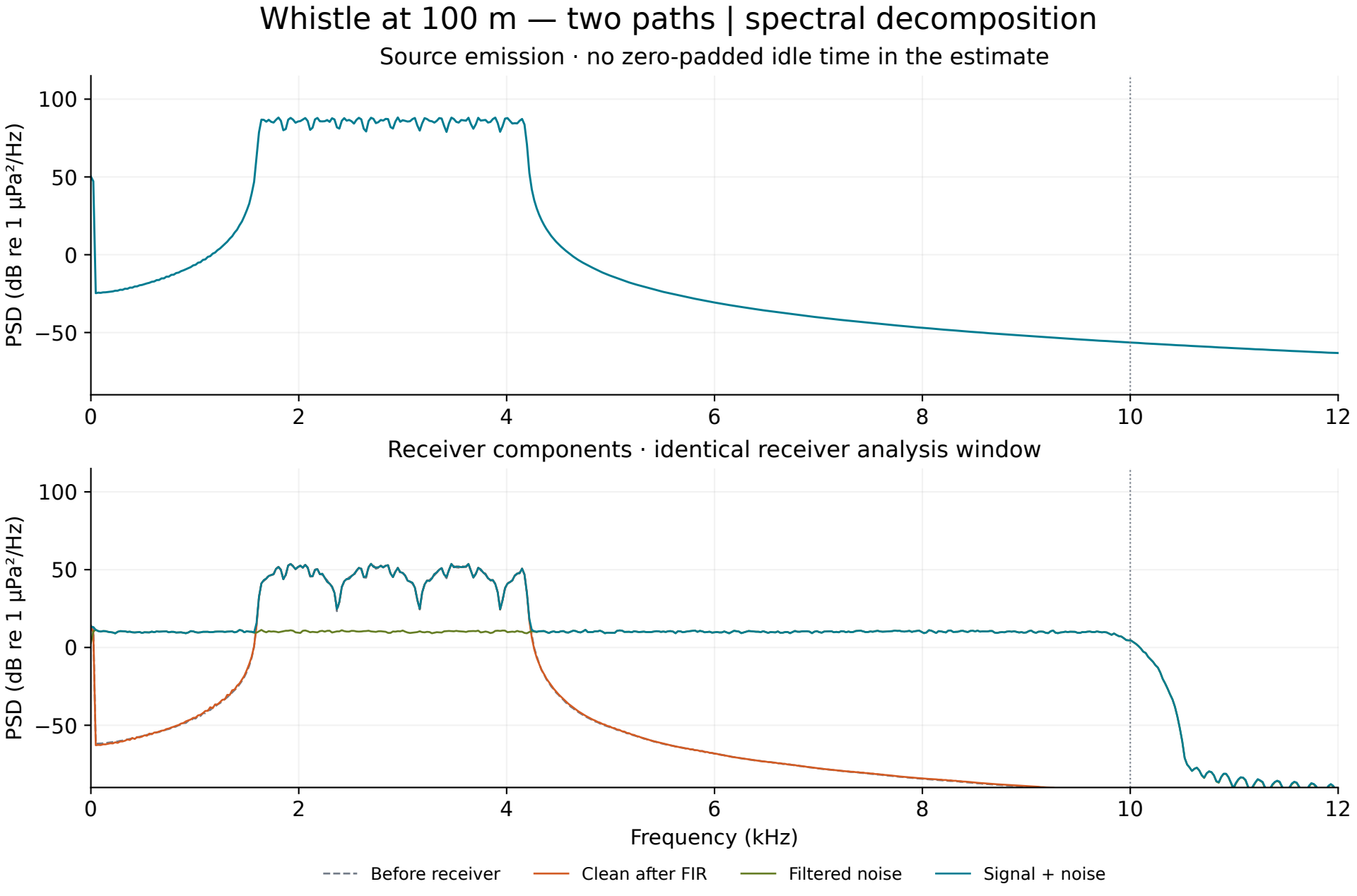
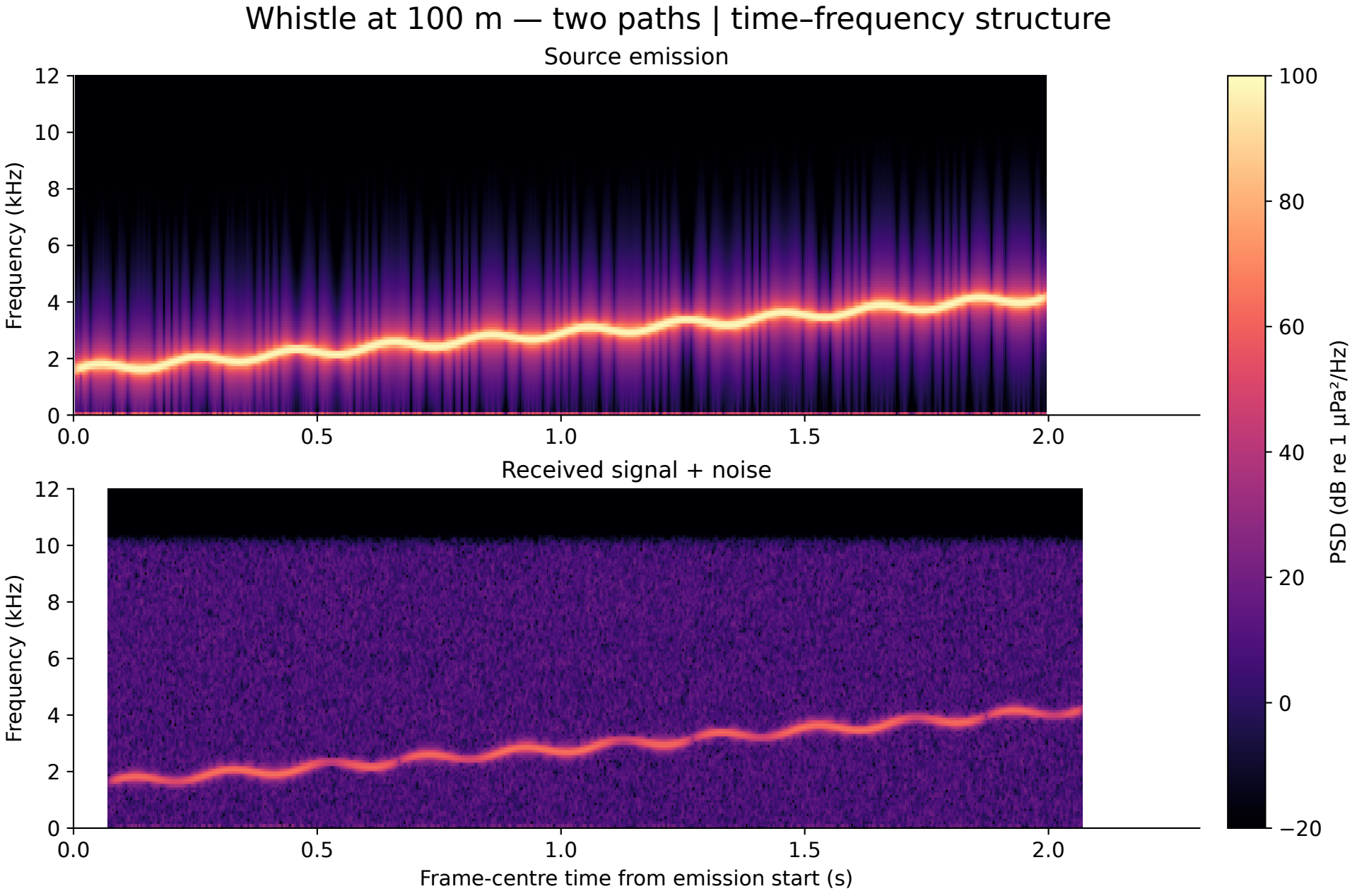
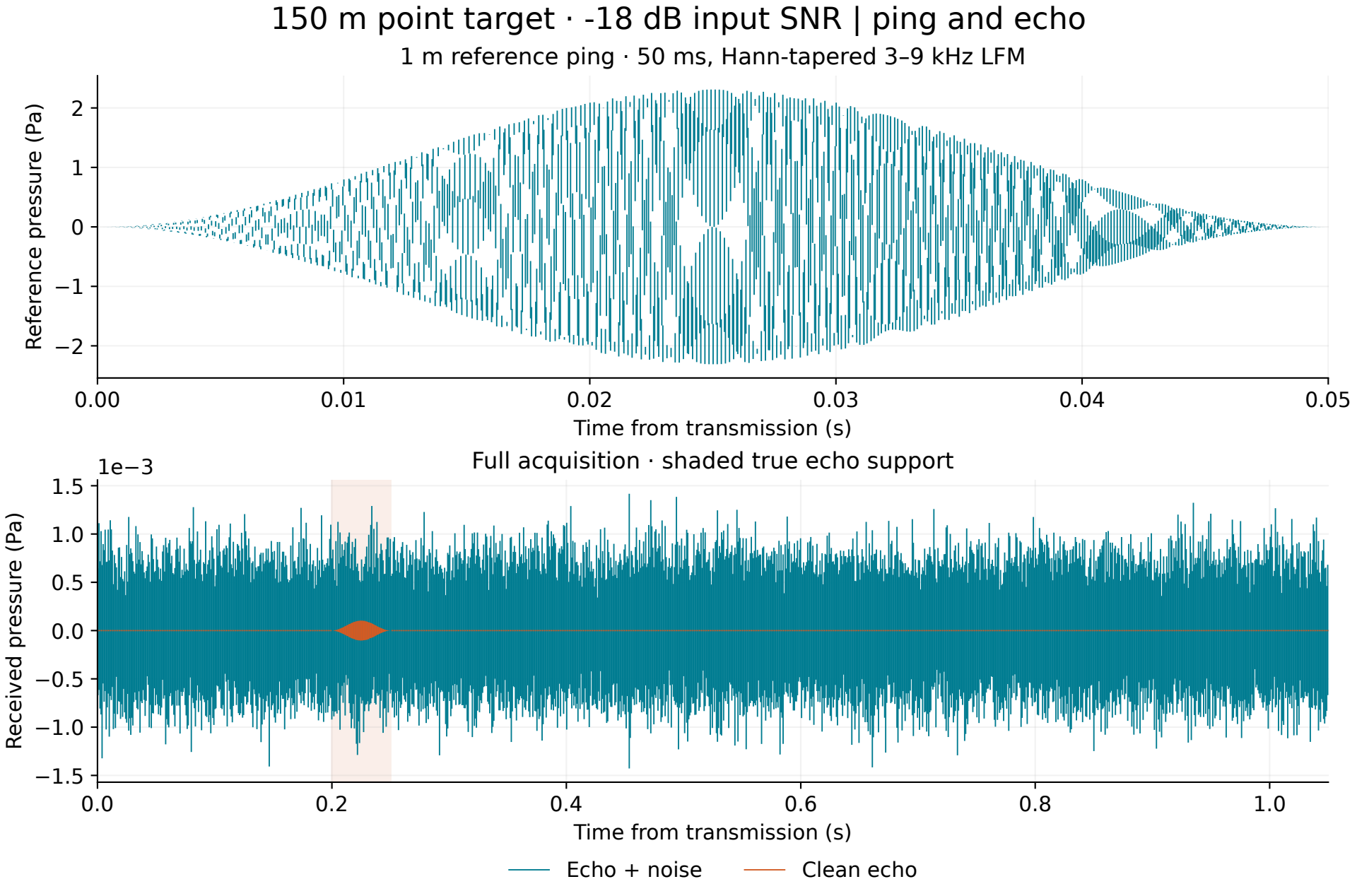
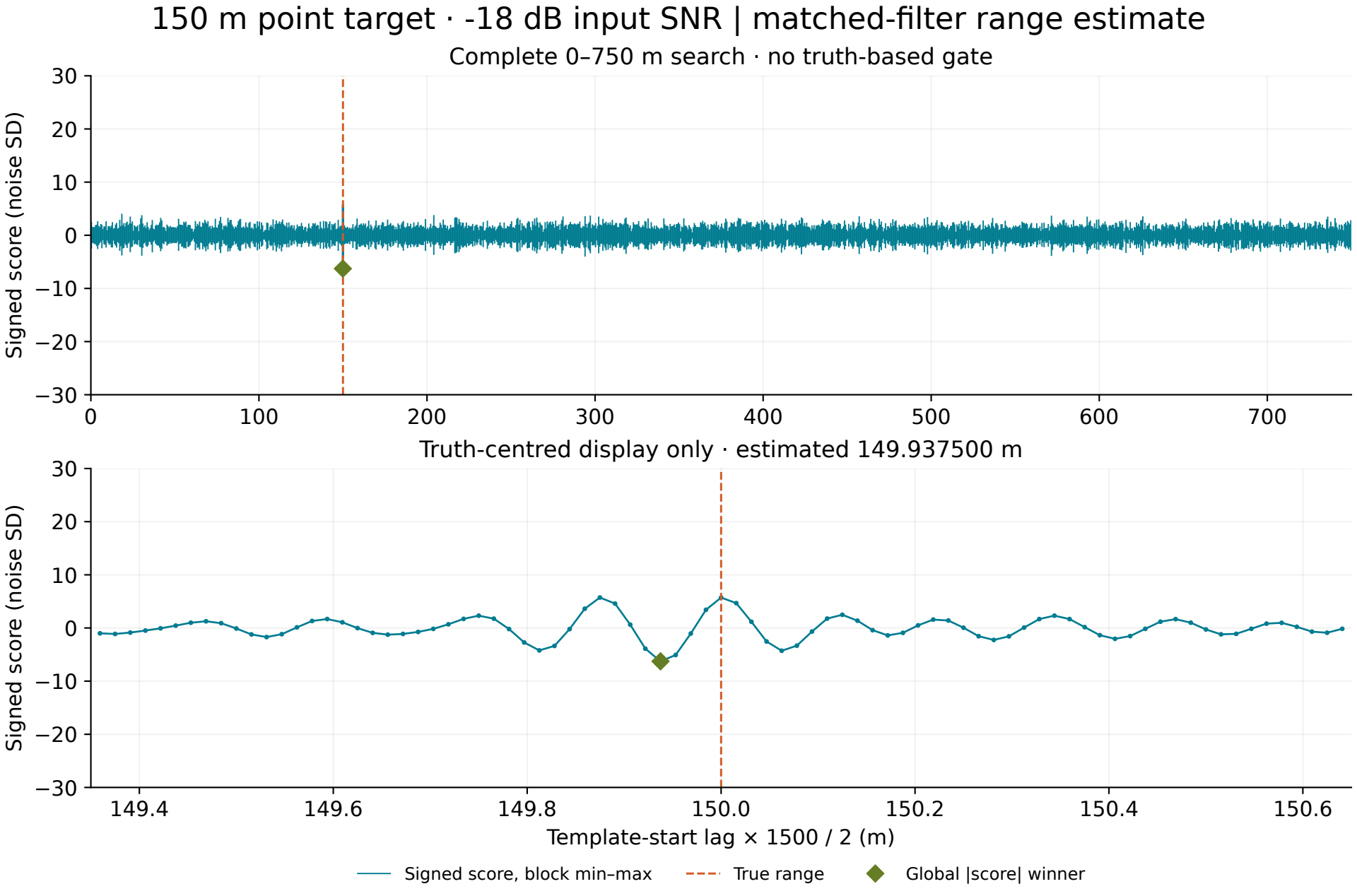
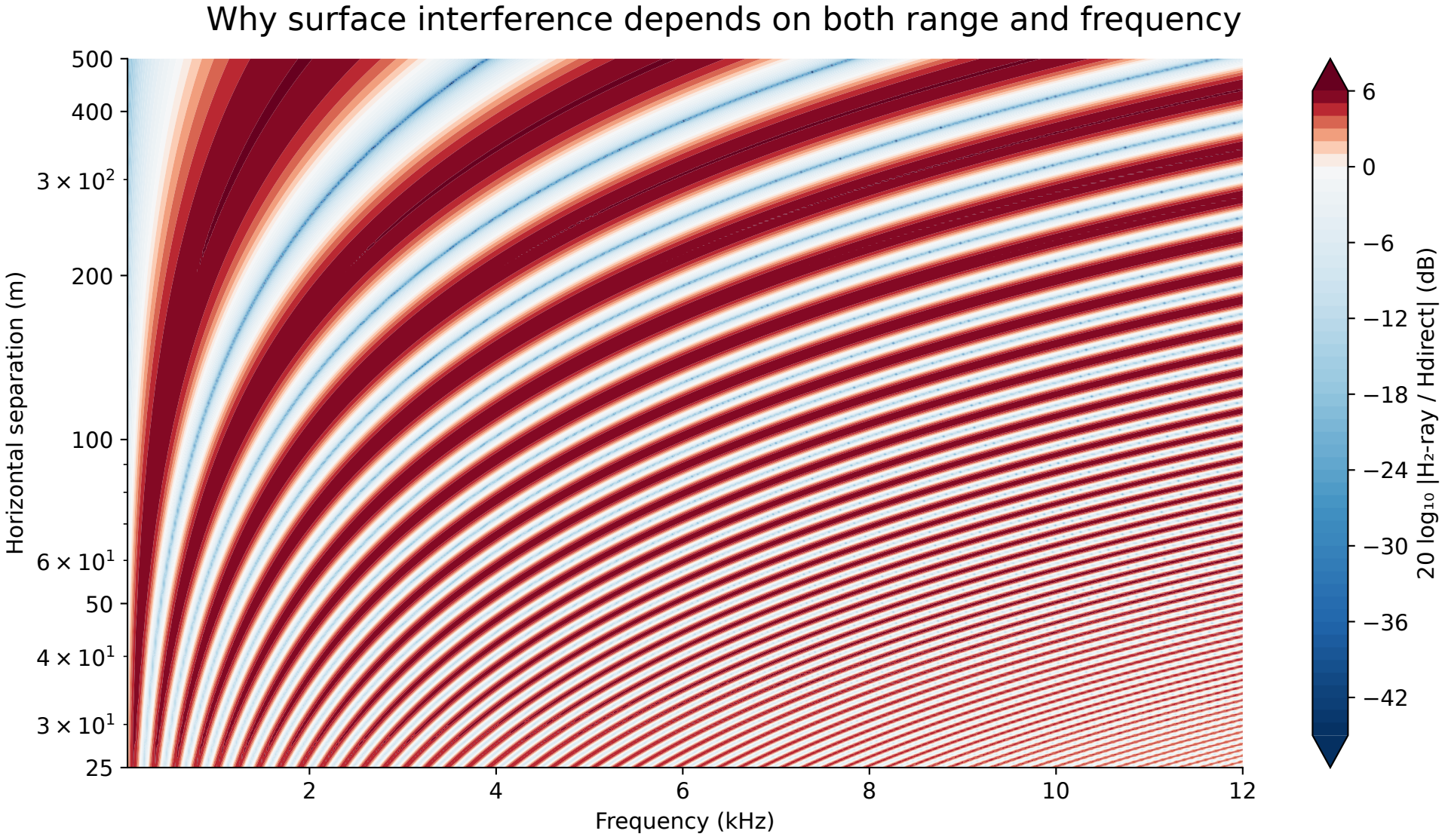

Whistle at 100 m — two paths | pressure histories
1 m reference source · RMS over unpadded emission
Source pressure (Pa)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.0
0.5
1.0
1.5
2.0
Receiver analysis [0.06538, 2.07467) s
1e−2
Received pressure (Pa)
3
2
1
0
−1
−2
−3
0.0
0.5
1.0
1.5
2.0
Time from emission start (s)
Signal + filtered noise
Clean after receiver

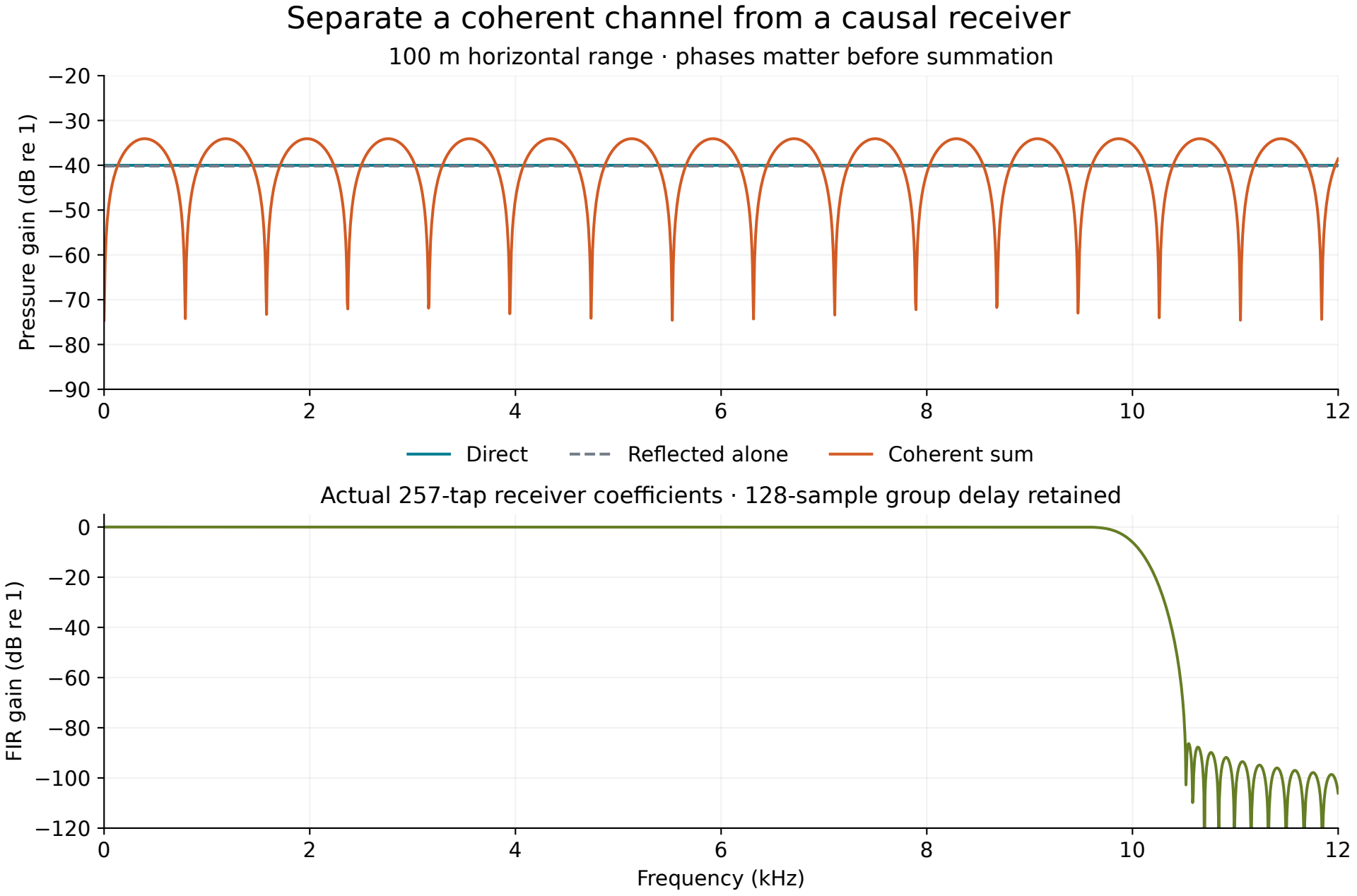

Whistle at 100 m — two paths | spectral decomposition
Source emission · no zero-padded idle time in the estimate
PSD (dB re 1 µPa²/Hz)
100
50
0
−50
0
2
4
6
8
10
12
Receiver components · identical receiver analysis window
PSD (dB re 1 µPa²/Hz)
100
50
0
−50
0
2
4
6
8
10
12
Frequency (kHz)
Before receiver
Clean after FIR
Filtered noise
Signal + noise

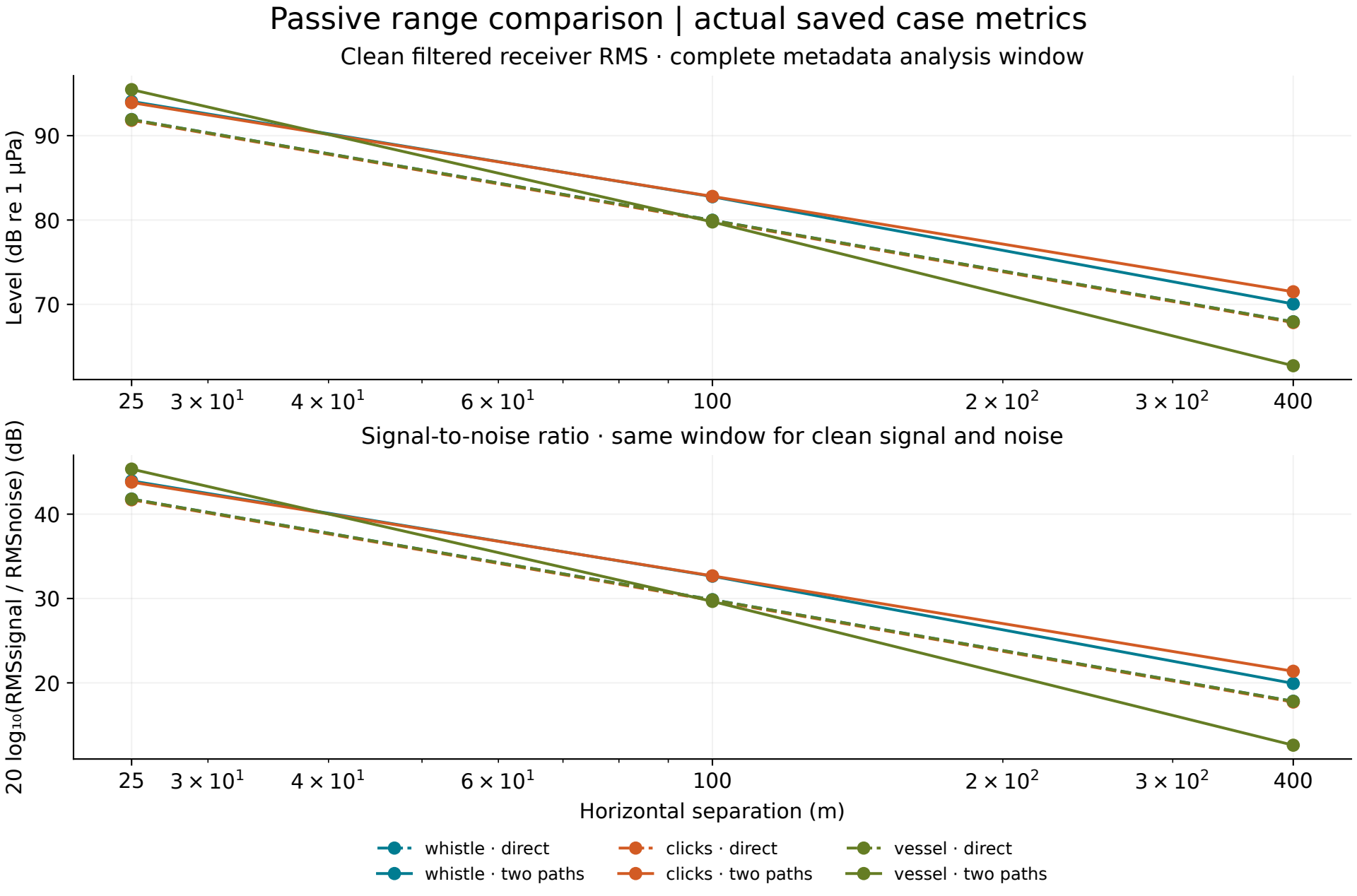

Whistle at 100 m — two paths | time–frequency structure
Source emission
Received signal + noise
Frequency (kHz)
Frequency (kHz)
0
2
4
6
8
10
12
0.0
0.5
1.0
1.5
2.0
Frame-centre time from emission start (s)
PSD (dB re 1 µPa²/Hz)
100
80
60
40
20
0
−20

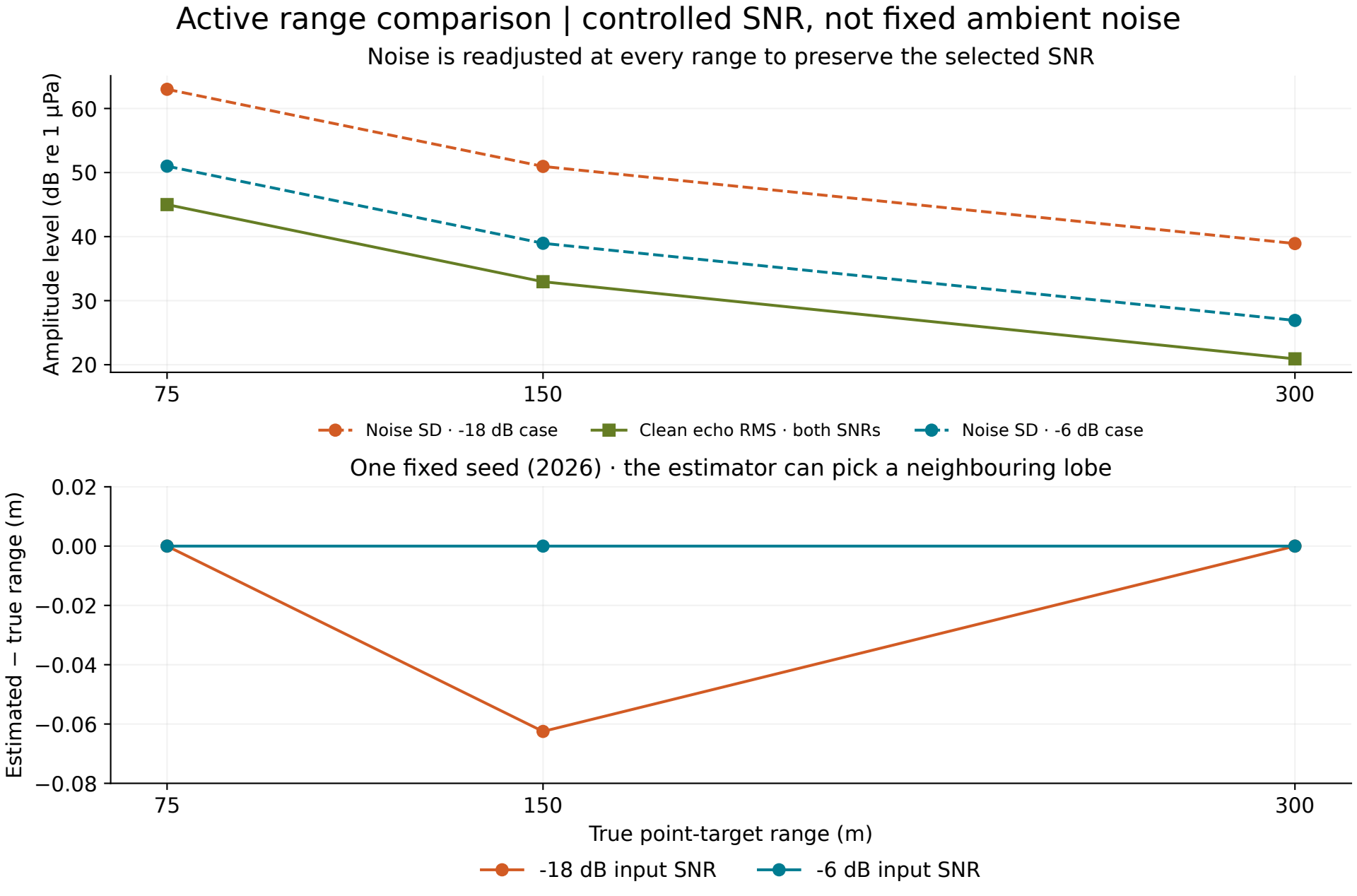

150 m point target · -18 dB input SNR | ping and echo
1 m reference ping · 50 ms, Hann-tapered 3–9 kHz LFM
Reference pressure (Pa)
2
1
0
−1
−2
0.00
0.01
0.02
0.03
0.04
0.05
Time from transmission (s)
Full acquisition · shaded true echo support
1e−3
Received pressure (Pa)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.0
0.2
0.4
0.6
0.8
1.0
Time from transmission (s)
Echo + noise
Clean echo

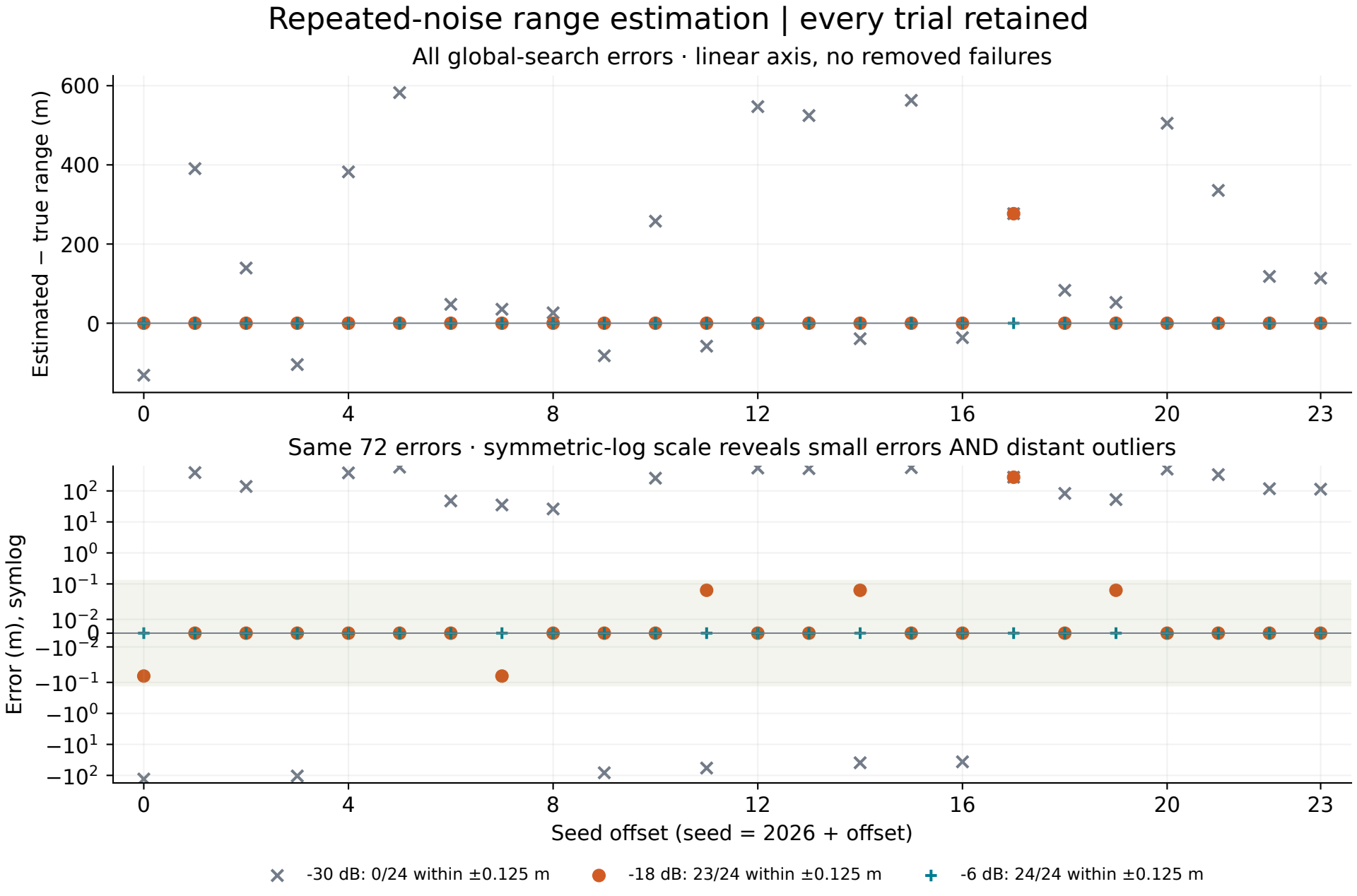

150 m point target · -18 dB input SNR | matched-filter range estimate
Complete 0–750 m search · no truth-based gate
Signed score (noise SD)
30
20
10
0
−10
−20
−30
0
100
200
300
400
500
600
700
Truth-centred display only · estimated 149.937500 m
Signed score (noise SD)
149.4
149.6
149.8
150.0
150.2
150.4
150.6
Template-start lag × 1500 / 2 (m)
Signed score, block min–max
True range
Global |score| winner


Why surface interference depends on both range and frequency
Horizontal separation (m)
500
400
3 × 10²
200
100
6 × 10¹
50
4 × 10¹
3 × 10¹
25
2
4
6
8
10
12
Frequency (kHz)
6
0
−6
−12
−18
−24
−30
−36
−42
20 log₁₀ |H₂-ray / Hdirect| (dB)


Separate a coherent channel from a causal receiver
100 m horizontal range · phases matter before summation
Pressure gain (dB re 1)
−20
−30
−40
−50
−60
−70
−80
−90
0
2
4
6
8
10
12
Direct
Reflected alone
Coherent sum
Actual 257-tap receiver coefficients · 128-sample group delay retained
FIR gain (dB re 1)
0
−20
−40
−60
−80
−100
−120
0
2
4
6
8
10
12
Frequency (kHz)


Passive range comparison | actual saved case metrics
Clean filtered receiver RMS · complete metadata analysis window
Level (dB re 1 µPa)
90
80
70
25
3 × 10^1
4 × 10^1
6 × 10^1
100
2 × 10^2
3 × 10^2
400
Signal-to-noise ratio · same window for clean signal and noise
20 log10(RMSsignal / RMSnoise) (dB)
40
30
20
Horizontal separation (m)
whistle · direct
whistle · two paths
clicks · direct
clicks · two paths
vessel · direct
vessel · two paths


Active range comparison | controlled SNR, not fixed ambient noise
Noise is readjusted at every range to preserve the selected SNR
Amplitude level (dB re 1 µPa)
60
50
40
30
20
75
150
300
Noise SD · -18 dB case
Clean echo RMS · both SNRs
Noise SD · -6 dB case
One fixed seed (2026) · the estimator can pick a neighbouring lobe
Estimated − true range (m)
0.02
0.00
−0.02
−0.04
−0.06
−0.08
75
150
300
True point-target range (m)
-18 dB input SNR
-6 dB input SNR


Repeated-noise range estimation | every trial retained
All global-search errors · linear axis, no removed failures
Estimated − true range (m)
600
400
200
0
0
4
8
12
16
20
23
Same 72 errors · symmetric-log scale reveals small errors AND distant outliers
Error (m), symlog
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
0
-10^{-2}
-10^{-1}
-10^{0}
-10^{1}
-10^{2}
Seed offset (seed = 2026 + offset)
-30 dB: 0/24 within ±0.125 m
-18 dB: 23/24 within ±0.125 m
-6 dB: 24/24 within ±0.125 m